YUANBO PANG

UC Berkeley EECS | Foundation Agents & Robotics
yuanbopang@berkeley.edu · (+1) 408-460-7526 · LinkedIn · GitHub

Research interests: RL for foundation-agent training, embodied agents, agent evaluation

Education

University of California, Berkeley — B.S. Electrical Engineering & Computer Sciences *Expected Dec 2026*
GPA: 3.811/4.0 Relevant coursework: CS 170, CS 168, CS 184, EECS 106A, CS 61B, CS 70

Stanford University (Summer Session) — CS 229: Machine Learning *Summer 2025*

Publications

Agents' Last Exam.
Yiyou Sun*, Xinyang Han*, Weichen Zhang*, **Yuanbo Pang***, et al., Dawn Song. *Under review at NeurIPS 2026.* arXiv:2606.05405.
(*Co-first author & core contributor; 310 authors total)

Research Experience

Research Assistant, Berkeley RDI / BAIR (Prof. Dawn Song) *Oct 2025 – Present*
Agents' Last Exam — co-first author on V1; leading two research tracks in V2

- **Foundation Agent / 3D spatial understanding.** Building training and evaluation environments in simulation (Blender, NVIDIA Isaac Lab) where agents actively move the viewpoint to explore a scene and judge physical state — 3D understanding through exploration, not inference from a fixed 2D view.
- **Ultra-long-horizon engineering.** Constructing tasks from large-scale production software that human teams measure in months to years, with a harness that evaluates *teams* of agents — coordination, division of labor, and context handoff at horizons far beyond single-agent benchmarks.
- For V1, built task/environment pipelines, computer-use and tool interfaces, scalable GCP execution, and programmatic evaluators producing verifiable rewards for RLVR; now run ALE evaluations with frontier model teams (OpenAI, Anthropic, Meta TBD Lab, Qwen, Seed), analyzing agent trajectories and reporting failure modes back to the labs.

Robotics Research, Prof. Masayoshi Tomizuka's Lab, UC Berkeley *2026 – Present*
Large-Scale Robot-Learning Data Benchmark — mentored by Zhuo Xu

- Building a benchmark that measures dataset quality by downstream finetune uplift — the Δ a dataset produces on each model's own native benchmark (LIBERO, SimplerEnv, CALVIN) — rather than by proxy metrics alone.
- Owning the agent-driven data layer: agents that audit annotation quality and auto-convert raw captures into the team's canonical, embodiment-tagged representation (egocentric human hand, parallel-jaw gripper, humanoid).
- Validating that cheap CPU-only quality signals predict that uplift, and that dataset rankings stay consistent across ACT, Diffusion Policy, OpenVLA, π_0, and GR00T.

URAP Research Assistant, Prof. Avideh Zakhor, UC Berkeley EECS *Spring 2026 – Present*

- Building a high-quality UAV-to-UAV dataset and PyTorch pipelines for zero-shot detection and multi-object tracking under high-speed ego-motion; integrating optical flow with YOLO and SAM2-style baselines. Targeting CVPR submission.

Engineering Experience

Founding Engineer, Equile — "Vercel for AI agent harnesses" *May 2026 – Present*

- Sole owner of the runtime and control plane: serverless hosting for agent harnesses (Claude Code, Codex, Cursor, Grok) — isolated per-session sandboxes, multi-tenant packing that cuts per-agent compute >20×, snapshot pause/resume, public SDK/CLI.

Agentic Tech Staff, Starbot *Nov 2025 – May 2026*

- Led the agentic stack for autonomous service robots: vision-language-action models, agentic planning and control, multi-robot coordination.
- Shipped as the CES 2026 demo system, deployed on the show floor 8 hours/day across coffee preparation, popcorn making, and front-desk reception; covered live by Japan's NHK News, demoed to 1,500+ attendees, and converted into 50+ inbound customer conversations.

Software Engineering Intern, Geopogo *Fall 2025*

- Shipped a production multimodal-generation workflow (React/TypeScript, FastAPI, Gemini, Firebase, CI/CD).

Selected Products & Awards

Ludus (co-founder): Dual-agent task marketplace with isolated execution, GitHub-based evaluation, and vote/Elo ranking.
Awards: HackMIT Modal Sponsor Prize (2025); Smartest AI Agent, HackMIT (2024); MLH Best Domain, HackDavis (2024).

Technical Skills

ML / Robotics / Sim: PyTorch, VLA, multi-object tracking, YOLO, SAM2, optical flow, Isaac Lab, Blender
Agent / Systems: Benchmarks, automated evaluators, tool-use & MCP, Docker, GCP, Modal, FastAPI, Postgres, Linux